A finite maximal tree of split pairs forces a last true instance on one of the two sides

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Editorial note. The counted unit is the article's own Lemma lem:sads-wf-finite-tree-forcing at source lines 1206-1215 together with the proof environment that begins on the next line (1216-1231); the pairing is the ordinary adjacent one. The author has left a commented-out earlier version of the same lemma immediately after the proof at source lines 1234 onwards; it is not author text for our purposes and is not retained.

Editorial note. The retained context at source lines 1149-1156 is the statement of the forcing question lemma, which the proof invokes twice, and it is counted by a sibling card of this lane; the retained context at source lines 1169-1183 and 1186-1192 is the author's own definitions of trees of split pairs, validity and satisfaction.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained.

Editorial note. No retained line contains a citation, so the card carries no bibliography; the only cross-references of the retained proof are the two author Cref calls to the forcing question lemma, whose statement is retained in the context.

Editorial note. The article's conservation theorems, its COH, WKL and SEM lemmas and its other SADS lemmas are not counted here.

Allowed standard prerequisites (editorial).

- Trees of split pairs, validity and satisfaction. A tree attached to a condition is a binary tree labelled by split pairs extending the condition, with successors extending the two components; a node is valid when along the left branches above it the ascending components lie inside U and along the right branches the descending components lie inside V; a node satisfies a pair of Sigma^0_1 formulas when the first holds of the ascending component at the node rank and the second of the descending component

at the number of ones, and a tree satisfies the pair when all its nodes do; the author's own definitions are retained at source lines 1169-1183 and 1186-1192.. Fixes the objects and the satisfaction property of the counted lemma, and supplies the validity notion used to pick the node whose failure of an extension contradicts maximality.

- The forcing question and its two-case decision. For a condition and two computations, the forcing question relates the two sides through a split pair, and the article proves that if the question fails then some extension forces divergence on one of the two sides, decidably so; retained with this card at source lines 1149-1156.. The proof uses it twice: once when the tree is empty, and once at the maximal valid node, to convert the failure of the forcing question into an extension forcing the negation of the next instance.

- Induction along the tree by L Delta^0_2. The scheme of induction for L Delta^0_2 formulas, available because the model satisfies collection for Sigma^0_2, allows a maximal valid element of a finite tree to be taken.. The proof selects a valid node with no valid descendant, which needs that scheme.

Lemma 0.1. *Let (σ_0, τ_0) be a condition, and let $e_0, e_1, x_0, x_1 \in M$. Then:*

- *If $(\sigma_0, \tau_0) \,?\vdash \Phi_{e_0}^{G_0 \oplus Y}(x_0) \downarrow \vee \Phi_{e_1}^{G_1 \oplus Y}(x_1) \downarrow$, then there exist an extension $(\sigma_1, \tau_1) \leqslant (\sigma_0, \tau_0)$ and some $i < 2$ such that $(\sigma_1, \tau_1) \Vdash_i \Phi_{e_i}^{G_i \oplus Y}(x_i) \downarrow$.*

- *If $(\sigma_0, \tau_0) \,?\nvdash \Phi_{e_0}^{G_0 \oplus Y}(x_0) \downarrow \vee \Phi_{e_1}^{G_1 \oplus Y}(x_1) \downarrow$, then there exist an extension $(\sigma_1, \tau_1) \leqslant (\sigma_0, \tau_0)$ and some $i < 2$ such that $(\sigma_1, \tau_1) \Vdash_i \Phi_{e_i}^{G_i \oplus Y}(x_i) \uparrow$.*

Furthermore, deciding which case holds and computing the extension (σ_1, τ_1) can be done Y'-computably.

Definition 0.2. *Fix a condition (σ, τ).*

A (σ, τ)-tree *is a binary tree T labelled by a family of split pairs $(\sigma_\rho, \tau_\rho)_{\rho \in T}$ such that for all $\rho \in T$:*

- *$\sigma \preceq \sigma_\rho$ and $\tau \preceq \tau_\rho$;*

- *For every $\mu \in T$, if $\rho \cdot 0 \preceq \mu$ then $\sigma_\rho \preceq \sigma_\mu$ and if $\rho \cdot 1 \preceq \mu$ then $\tau_\rho \preceq \tau_\mu$;*

A node ρ is said to be valid *if for every $\mu \in T$ such that $\mu \cdot 0 \preceq \rho$, $\sigma_\mu \subseteq U$ and for every $\mu \in T$ such that $\mu \cdot 1 \preceq \rho$, $\tau_\mu \subseteq V$. Being valid is a $\Delta_2^0(Y)$ condition.*

Definition 0.3. *Let T a (σ, τ)-tree*

- *A node (σ_ρ, τ_ρ) in T* satisfies *a pair of $\Sigma_1^0(Y)$-formulas $(\varphi_0(G_0, x), \varphi_1(G_1, x))$ if $\varphi_0(\sigma, |\rho| - \operatorname{card} \rho)$ and $\varphi_1(\tau, \operatorname{card} \rho)$ holds (where $\operatorname{card} \rho$ is the cardinal of ρ seen as a set).*

- *T* satisfies *$(\varphi_0(G_0, x), \varphi_1(G_1, x))$ if every node of T satisfies it.*

Lemma 0.4. *Let (σ, τ) be a condition, $(\varphi_0(G_0, x), \varphi_1(G_1, x))$ be a pair of $\Sigma^0_1(Y)$ formulas, and T be an M-finite maximal (σ, τ)-tree satisfying $(\varphi_0(G_0, x), \varphi_1(G_1, x))$. Then there is an extension $(\hat{\sigma}, \hat{\tau}) \leqslant (\sigma, \tau)$ and some $p, q \in M \cup \{-1\}$ such that the following holds:*

- $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \varphi_0(G_0, p)$ *and* $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \varphi_1(G_1, q)$
- $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \neg\varphi_0(G_1, p+1)$ *or* $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \neg\varphi_1(G_1, q+1)$.

With the convention that $(\hat{\sigma}, \hat{\tau}) \Vdash_i \varphi_i(G^i, -1)$ always holds.

Proof. If T is empty, then $(\sigma, \tau) ?\not\vdash \varphi_0(G_0, 0) \vee \varphi_1(G_1, 0)$ as otherwise the corresponding split pair could be added in position ϵ in T, contradicting its maximality. Thus, by Lemma 0.1, there exists some extension $(\hat{\sigma}, \hat{\tau}) \leqslant (\sigma, \tau)$ and some side $i < 2$ such that $(\hat{\sigma}, \hat{\tau}) \Vdash_i \neg\varphi_i(G_i, 0)$. Thus, we can assume that T is not empty.

Consider a valid $\rho \in T$ having no valid descendant in T, this is possible since the set of valid element of T is non empty because $(\sigma_\epsilon, \tau_\epsilon)$ is a vacuously valid, so we can take a maximal valid element of T using $\mathsf{L}\Delta^0_2$ (which we have by $\mathsf{B}\Sigma^0_2$).

Since (σ_ρ, τ_ρ) is a split pair, we have two cases:

Case 1: $\sigma_\rho \subseteq U$. In that case $\rho \cdot 0 \notin T$ by maximality of ρ. Then take $p = |\rho| - \operatorname{card}\rho$, $q = \operatorname{card}\rho - 1$ and $(\sigma', \tau') = (\sigma_\rho, \tau_{\rho\restriction(i_0+1)})$ with i_0 the biggest index $i < |\rho|$ such that $\rho(i) = 1$ (if no such index exists we take τ instead of $\tau_{\rho\restriction(i_0+1)}$), the pair (σ', τ') is a valid condition by definition of validity and the hypothesis $\sigma_\rho \subseteq U$. $(\sigma', \tau') \Vdash_0 \varphi_0(G_0, p)$ and $(\sigma', \tau') \Vdash_1 \varphi_1(G_1, q)$ because T satisfies $(\varphi_0(G_0, x), \varphi_1(G_1, x))$.

We have $(\sigma', \tau') ?\not\vdash \varphi_0(G_0, p+1) \vee \varphi_1(G_1, q+1)$ as otherwise the corresponding split pair could be added in position $\rho \cdot 0$ in T, contradicting its maximality. Hence, by Lemma 0.1, there exists some extension $(\hat{\sigma}, \hat{\tau})$ and such that either $(\hat{\sigma}, \hat{\tau}) \Vdash_0 \neg\varphi_0(G_0, p+1)$ or $(\hat{\sigma}, \hat{\tau}) \Vdash_1 \neg\varphi_1(G_1, q+1)$ holds.

Case 2: The case $\tau_\rho \subseteq V$ can be treated similarly.

□